CAPSTONE PROJECT

Project Title: Hospital Patient Data Analysis

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1. Problem Statement

This project focuses on analyzing hospital patient data to gain insights into disease patterns, doctor workload, and treatment costs. The dataset includes details such as patient demographics, diagnosed diseases, assigned doctors, and treatment expenses.

The main objective is to understand which diseases are most common, which doctors are handling the highest number of patients, and how treatment costs vary across different cases. These insights can help hospital management make better decisions regarding resource allocation, cost management, and patient care improvement.

2. Solution/Features

The project was carried out using a combination of SQL, Google Sheets, and Python to perform data analysis and visualization.

SQL was used to create the database structure, insert patient data, and execute queries to extract meaningful insights such as disease frequency, average treatment cost, and doctor workload.

Google Sheets was used to create pivot tables, which helped in summarizing the data in a simple and visual format. This made it easier to identify patterns and compare different categories.

Python, along with libraries like Pandas and Matplotlib, was used to perform data analysis and generate visualizations such as pie charts and bar charts. These visual representations helped in better understanding the distribution of diseases and workload among doctors.

Overall, the combination of these tools allowed for efficient data processing, analysis, and visualization.

3. Screenshots

```
(2,30,'Female','Fever','Dr Patil',2000),
(3,55,'Male','Heart Disease','Dr Shah',20000),
(4,22,'Female','Flu','Dr Kulkarni',1500),
(5,40,'Male','Diabetes','Dr Shah',4500);
SELECT * FROM patients;

SELECT * FROM patients;
SELECT disease, COUNT(*) AS patient_count
```

Result Grid

disease	patient_count
Diabetes	2
Fever	1
Heart Disease	1
Flu	1

Output

#	Time	Action	Message	Duration / Fetch
6	16:04:00	INSERT INTO patients VALUES (1,45,'Male','Diabetes','Dr Shah',5000), (2,30,'Fem...	Error Code: 1146. Table 'hospital_db.patients' doesn't exist	0.000 sec
7	16:04:33	CREATE TABLE patients (patient_id INTEGER, age INTEGER, gender TE...	0 row(s) affected	0.047 sec
8	16:04:47	INSERT INTO patients VALUES (1,45,'Male','Diabetes','Dr Shah',5000), (2,30,'Fem...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	16:05:03	SELECT * FROM patients LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	16:05:56	SELECT disease, COUNT(*) AS patient_count FROM patients GROUP BY disease ...	4 row(s) returned	0.000 sec / 0.000 sec

DESEASE AND PATIENT COUNT

```
(5,40,'Male','Diabetes','Dr Shah',4500);
SELECT * FROM patients;

SELECT * FROM patients;
SELECT disease, COUNT(*) AS patient_count
FROM patients
GROUP BY disease;
```

avg_cost
6600.0000

#	Time	Action	Message	Duration / Fetch
7	16:04:33	CREATE TABLE patients (patient_id INTEGER, age INTEGER, gender TE...	0 row(s) affected	0.047 sec
8	16:04:47	INSERT INTO patients VALUES (1,45,'Male','Diabetes','Dr Shah',5000), (2,30,'Fem...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	16:05:03	SELECT * FROM patients LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	16:05:56	SELECT disease, COUNT(*) AS patient_count FROM patients GROUP BY disease ...	4 row(s) returned	0.000 sec / 0.000 sec
11	16:06:19	SELECT AVG(treatment_cost) AS avg_cost FROM patients LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

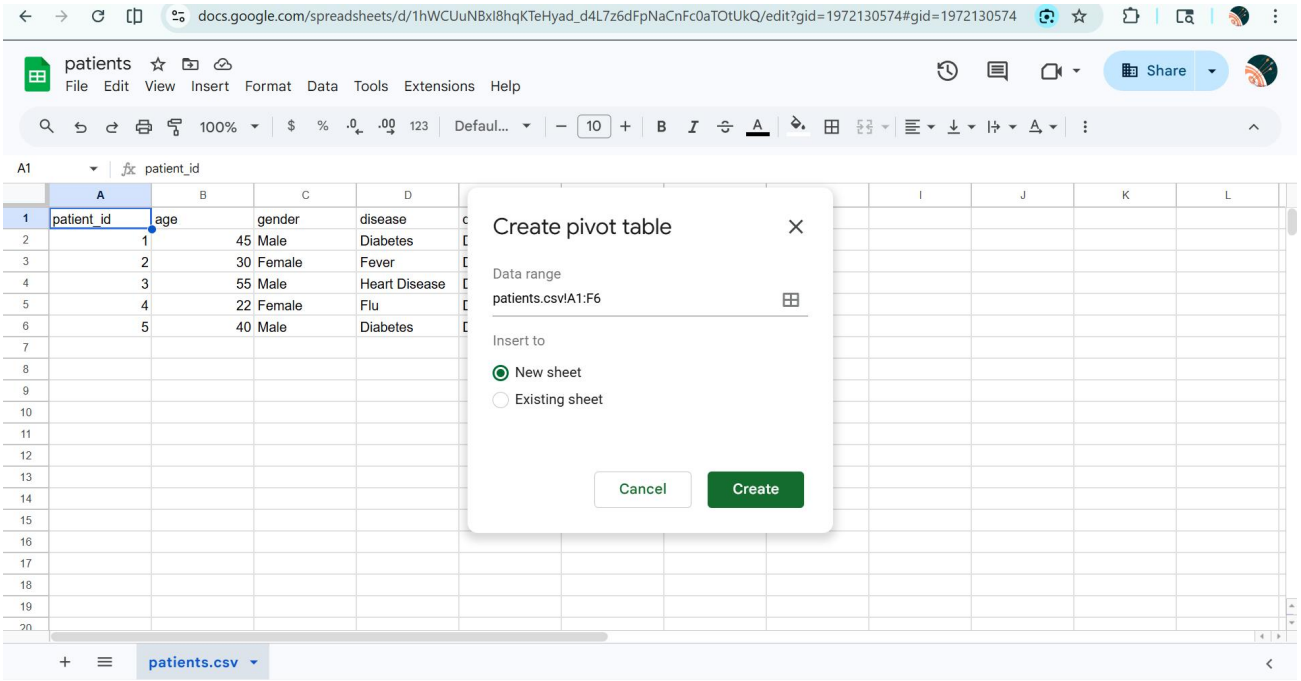
AVERAGE COST QUERY TABLE

```
SELECT disease, COUNT(*) AS patient_count
FROM patients
GROUP BY disease;

SELECT AVG(treatment_cost) AS avg_cost
FROM patients;

SELECT doctor, COUNT(*) AS total_patients
```

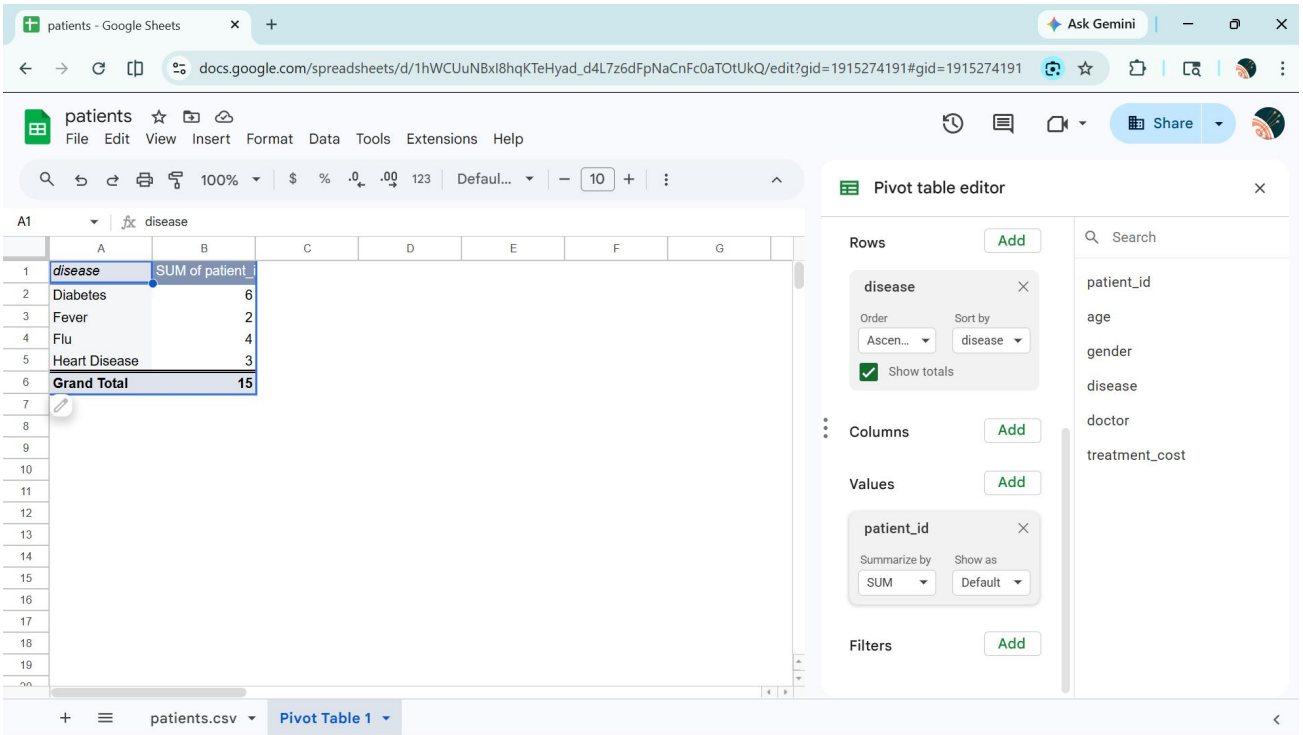
doctor	total_patients
Dr Shah	3
Dr Patil	1
Dr Kulkarni	1

#	Time	Action	Message	Duration / Fetch
8	16:04:47	INSERT INTO patients VALUES (1,45,'Male','Diabetes','Dr Shah',5000), (2,30,'Fem...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	16:05:03	SELECT * FROM patients LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	16:05:56	SELECT disease, COUNT(*) AS patient_count FROM patients GROUP BY disease ...	4 row(s) returned	0.000 sec / 0.000 sec
11	16:06:19	SELECT AVG(treatment_cost) AS avg_cost FROM patients LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
12	16:06:41	SELECT doctor, COUNT(*) AS total_patients FROM patients GROUP BY doctor O...	3 row(s) returned	0.000 sec / 0.000 sec

TOTAL PATIENTS QUERY SCREENSHOT



CREATING PIVOT TABLE OPTIONS



PIVOT TABLE CREATED SUCCESFULLY

4. Tech Stack

The project was developed using the following technologies:

- SQL (MySQL/SQLite): Used for database creation, data insertion, and performing analytical queries.

- Python (Pandas, Matplotlib): Used for data manipulation, analysis, and visualization.
- Google Sheets: Used for creating pivot tables and summarizing data in a structured format.

5. Unique Points

This project stands out from standard data analysis reports due to the following unique and practical features:

● Net Job Impact Calculation:
Instead of only highlighting the number of jobs replaced by AI, this project introduces a more balanced metric called "Net Job Impact." This is calculated by subtracting the number of jobs lost from the number of jobs created. This approach provides a more realistic and meaningful understanding of how AI is transforming the workforce, rather than presenting a one-sided perspective.

● Training vs Productivity Analysis:
A dedicated scatter plot visualization was created to analyze the relationship between AI training and employee productivity. By comparing "AI Training Hours" with "Productivity Change," the analysis clearly demonstrates whether investing time in training leads to measurable performance improvements. This adds a decision-making perspective to the project.

● Data-Driven Insights:
The project focuses on extracting actionable insights rather than just presenting raw data. Each analysis step is designed to answer specific business questions, such as identifying patterns, trends, and performance indicators, making the findings more useful for real-world applications.

● Clear and Simple Visualization:
The use of charts such as bar graphs, pie charts, and scatter plots ensures that complex data is presented in an easy-to-understand format. This improves readability and helps even non-technical users quickly grasp key insights.

● Professional Design and Presentation:
The project follows a clean and structured layout with well-organized sections, making it easy to navigate. Visual elements are designed to be clear and impactful, giving the overall project a professional and polished appearance suitable for academic and practical use.

● Multi-Tool Integration:
This project effectively combines multiple tools such as SQL, Python, and spreadsheet-based analysis. This integration demonstrates the ability to handle data across different platforms and showcases a complete end-to-end data analysis workflow.

6. Future Improvements

There are several ways this project can be further enhanced::-

- - Incorporating a larger dataset with more patient records to improve the accuracy and depth of analysis.
- - Developing an interactive dashboard using tools like Power BI or Tableau for better data visualization and real-time insights.
- - Applying advanced data analysis techniques such as predictive modeling to forecast disease trends and treatment costs.
- - Including additional parameters such as patient history, treatment duration, and recovery rates for more comprehensive analysis.